

LG 1: An Introduction to Genetics- The Roles of Genetics in Society and Principles of Heredity

The word “**genetics**” derives from ancient Greek; “genetikos”, which comes from “genesis” meaning “origin”. **Genetics** is a branch of biology that deals with the study of:

- the molecular structure and function of genes
- patterns of inheritance from parent to offspring (i.e., heredity)
- gene distribution
- Variation in populations.

Medical genetics is a branch of medicine that deals with the inheritance of diseases in families (using pedigree analysis) and diagnosis and treatment of diseases due to single gene mutations, chromosome abnormalities, and multifactorial disorders. (Multifactorial disorders are those associated with the effects of multiple genes in combination with lifestyle and environmental factors, and includes conditions like diabetes, obesity etc.)

The importance of genetics: Genetics has close connections with our everyday lives; it relates to medicine, agriculture, industry, technology, and ethics. We all possess genes that influence our lives. They affect our height, weight, hair color, and skin pigmentation. They influence our susceptibility to many diseases and disorders and even contribute to our intelligence, talents and personality.

Today, the major crops and animals used in agriculture have undergone extensive genetic alteration to greatly increase their yields and provide many desirable traits, such as resistance to drought, diseases and pest, in addition to special nutritional qualities (i.e., substances of commercial value). Growth hormones, insulin, vaccines, interferons, erythropoietin and clotting factors are now produced commercially by genetically engineered bacteria.

An Example of genetically modified organisms (GMOs) is “golden rice”. Most people around the world use rice as a dietary staple. Ordinary rice doesn’t contain much (if any) vitamin A. Vitamin A deficiency can cause preventable blindness in children and has many other effects. Golden rice is modified to have 23 times more vitamin A than ordinary rice and contains twice as much Fe. Golden rice has been very useful in aiding developing countries.

Edible vaccines have also been placed in GM crops. Genetic modification can be used to make crops larger and more attractive-looking. Another example of the use of GMOs is modified tomatoes with delayed ripening times so that they last longer.

Genetics also plays a critical role in medicine. Physicians recognize that many diseases and disorders have a hereditary background, including well-known disorders such as sickle-cell anemia, hemophilia, and Huntington disease as well as many common diseases such as asthma, diabetes, and hypertension. Advances in molecular genetics have allowed the development of many diagnostic tests. Gene therapy is the replacement of a missing gene product by inserting a normal gene into the somatic cells of patients in order to correct genetic diseases and has now been carried out on thousands of patients.

*Sickle cell anemia is a group of disorders that affect hemoglobin; patients have an abnormal form of hemoglobin called *hemoglobin S*, which distorts the RBCs into the shape of a sickle.

*Huntington’s disease is a progressive brain condition that caused uncontrolled movements and emotional and thinking problems. This condition causes the progressive breakdown of nerve cells in the brain.

*DNA chips, also called microarrays, are pieces of glass that contain many wells, with each well containing a specific gene. DNA chips are used to detect abnormal genes.

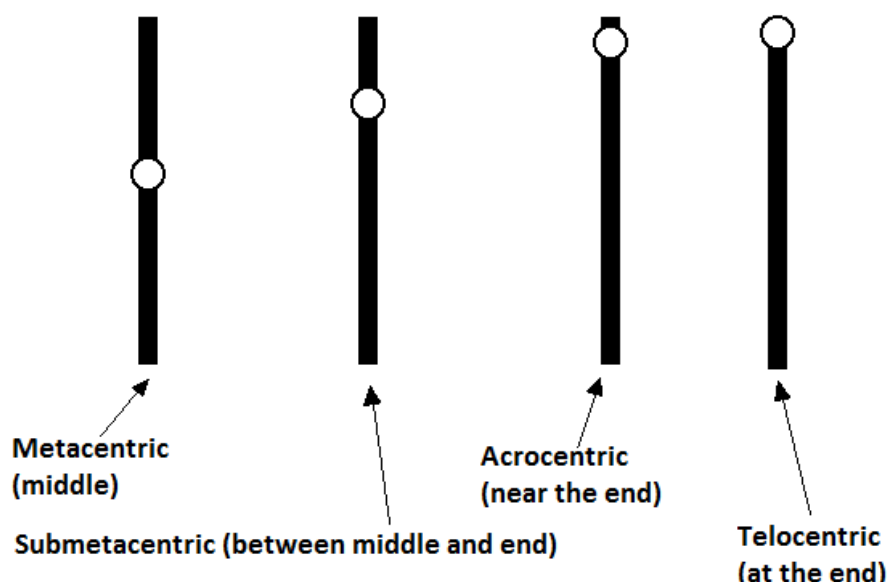
Terminology:

1. **Gene:** The basic unit of heredity in the chromosomes; a sequence of nucleotides in a DNA molecule that performs a specific function, such as coding for an RNA molecule or a polypeptide.
2. **Genome:** The complete set of chromosomes with their associated genes. The genome of an organism is its entire genetic information, including the product-coding and non-coding parts (e.g., regulatory elements and promoters).
3. **Alleles:** Two or more different forms of a gene. Alleles occupy the same position (locus) on homologous chromosomes and are separated from each other at meiosis.
4. **Genotype:** the sum total of all the genes present in an individual.
5. **Phenotype:** Observable (or detectable) characteristics of an organism resulting from interactions between the genotype and environment.
6. **Heredity:** The transmission of characteristics from parents to offspring. (Note: genetics is a lot wider than just heredity)
7. **Locus:** In genetics, the locus is the position of a gene in chromosomes. For any given locus, there may be a number of possible alleles.
8. **Karyotype:** The general appearance of the chromosomes of an organism with regard to number, size and shape, centromere location, and banding patterns. Karyotypes can be used to chromosomal mutations and for diagnosis.



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After staining, a chromosome shows a particular banding pattern



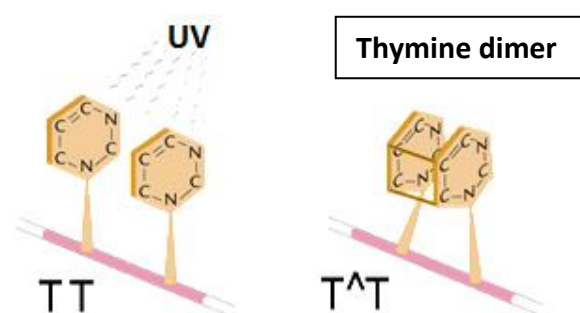
Note: Karyotypes show the appearance of chromosomes during metaphase, where they are condensed and easily distinguishable)

9. **Mutation:** A permanent change in a gene from one allelic form to another; an inheritable change in the nucleotide sequence of chromosomes.

10. **Mutant:** An organism carrying a gene that has undergone a mutation.

11. **Mutagen:** A chemical (such as nitrous acid, 5-bromouracil and Benzopyrene), physical (such as ionizing radiation like x-rays and UV light) or biological (viruses and bacteria) agent that increases the rate of mutation.

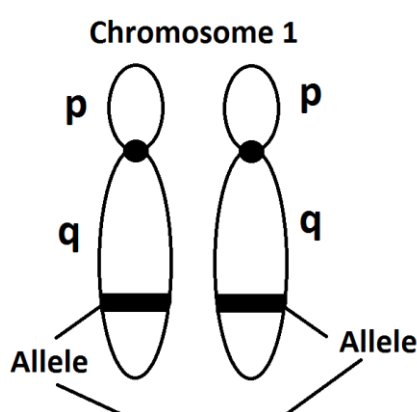
Note: UV light can cause thymine dimers.



12. **Teratogen:** an agent that can produce a permanent alteration in structure or function of an organism after exposure during embryonic or fetal life. They can cause congenital abnormalities and spontaneous abortions. Teratogens include environmental factors, medication, abuse of drugs, and chemicals. For example, the consumption of alcohol (the teratogen in this case) by a pregnant mother causes the fetus to develop Fetal Alcohol Syndrome (FAS). Another example is how taking tetracyclines during pregnancy causes the child to have permanently yellow teeth
Note: One way teratogens are different from mutagens is that they affect the fetus or embryo much more than the mother.

13. **Recessive allele:** An allele whose phenotypic effect is masked in a heterozygous by that of another, dominant allele.

14. **Dominant allele:** An allele whose phenotypic effect is the same in both heterozygous and homozygous conditions.



Say for example that each allele produces 50% of the gene's product.
Say one allele is mutated, resulting in the production of an abnormal product. If 50% of the normal product is enough for the disease this mutation would otherwise cause not to develop, the disease will be

15. Haploid and Diploid: The number of chromosomes in gametes is referred to as haploid (single set), and the number in somatic cells as diploid (double sets). Cells that have more than two sets of chromosomes are known as polyploidy (many sets).

Extra terminology:

- Recombination: taking a part of DNA and putting it in another piece of DNA
- Transgenic: moving one gene from a place to another

LG 2: Extending Mendelian Genetics

Many geneticists have extended Mendelian principles to patterns of inheritance more complex than that describes by Mendel.

Allele interactions (incomplete dominance) (50/50 expressed): A type of inheritance in which the offspring shows characteristics intermediate between two extreme parental features (i.e., neither allele is fully dominant). This phenomenon is known as incomplete dominance.

For example, a cross between a homozygous red flowering snapdragon (RR) and a homozygous white flowering snapdragon (R'R') produces heterozygous offspring with pink flowers. This is not an example that supports the blending theory of inheritance because, when F1 is allowed to self-pollinate, the F2 generation has a phenotypic ratio of 1 red, 2pink, 1white flowered plant (and thus the parental phenotypes reappeared in the F2). The cross:

P₁: RR & R'R'

F₂: RR' x RR' = RR + 2RR' + R'R'

F₁: RR x R'R' = RR' (100% pink)

1(red): 2(pink): 1(white)

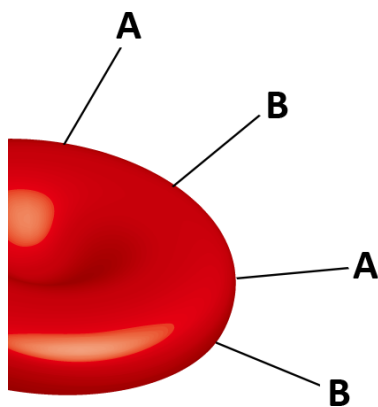
*Other examples of incomplete dominance include the inheritance of hypercholesterolemia, Tay-Sachs disease and sickle cell anemia.

Co-dominance (both alleles 100% expressed): Alleles may act in a co-dominant manner in which both alleles are apparently fully expressed in the phenotype of a heterozygote. An individual with the blood type AB is exhibiting co-dominant characteristics in which two different types of glycoprotein appear on the RBCs.

$I^A I^B \times I^A I^B$
 $I^A I^A, I^A I^B, I^A I^B, I^B I^B$
 1: 2: 1

Multiple alleles (several alleles for one trait) (not in an individual; in a population/species): Sometimes, there are more than two alleles for a given chromosomal locus, in which case a trait is controlled by multiple alleles. In rabbits, there are four alleles for coat color: $C > c^{ch} > c^h > c^a$ (Agouti, chinchilla, hemalaya, and albino, respectively).

The ABO blood groups in humans are controlled by three alleles and there are four possible phenotypes; A, B, AB, and O. A person's red blood cells may be coated with one glycoprotein surface antigen (type A or B), with both (type AB) or with neither (type O). Matching compatible blood groups is critical for blood transfusions. A person with type A has anti-B antibodies and one with type B has anti-A antibodies, and blood cells clump, or agglutinate, when they are mixed with antibodies against the antigens they bear. This agglutination can cause destruction of blood cells, tissue breakdown, fever and death. People with type AB blood, having neither anti-A nor anti-B antibodies, can accept blood of any type while people with type O blood, having no A or B antigens on their blood cells, can donate to anyone.



***AB blood type**

I^A = allele for A
 I^B = allele for B
 I^0/i = allele for O

A: $I^A I^A$ or $I^A i$
B: $I^B I^B$ or $I^B i$
AB: $I^A I^B$
O: ii

*Antigen A → N-Acetylgalactosamine is the terminal sugar; Antigen B → galactose is the terminal sugar.

Epistasis: A phenomenon in which a difference or mutation in one gene inhibits or masks the effects of another gene. A prime example of this is the “Bombay phenotype”, named after the city where it was first

discovered. Normally, a child with blood type O cannot result from an AB parent. However, an O child can result from an AB parent due to epistasis. This occurs due to a mutation in a gene that codes for the H-antigen, also called the H-substrate, a precursor for the ABO blood group antigen. Individuals with the homozygous recessive genotype, hh, have a deficiency in H-substrate (this is known as the Bombay phenotype), and since their precursors can't be produced, the antigens A and B can't be produced either despite the person's genotype having alleles for them, resulting in a blood type of O.

Pleiotropy (a gene that controls many traits): the term pleiotropy is used to describe a gene that affects more than one characteristic of the individual. Individuals with Marfan syndrome tend to be tall and thin with long legs, arms, and fingers. They are nearsighted, have ectopic lens (i.e., their lens aren't in the right place) and the wall of their aorta is weak, causing it to enlarge and eventually to split. All of these effects are caused by an inability of a gene to produce normal connective tissue (the gene is located on chromosome 15 and codes for a collagen fiber called *fibrin*, which has an effect on the structural integrity of bones and blood vessels).

*another example of pleiotropy is cystic fibrosis

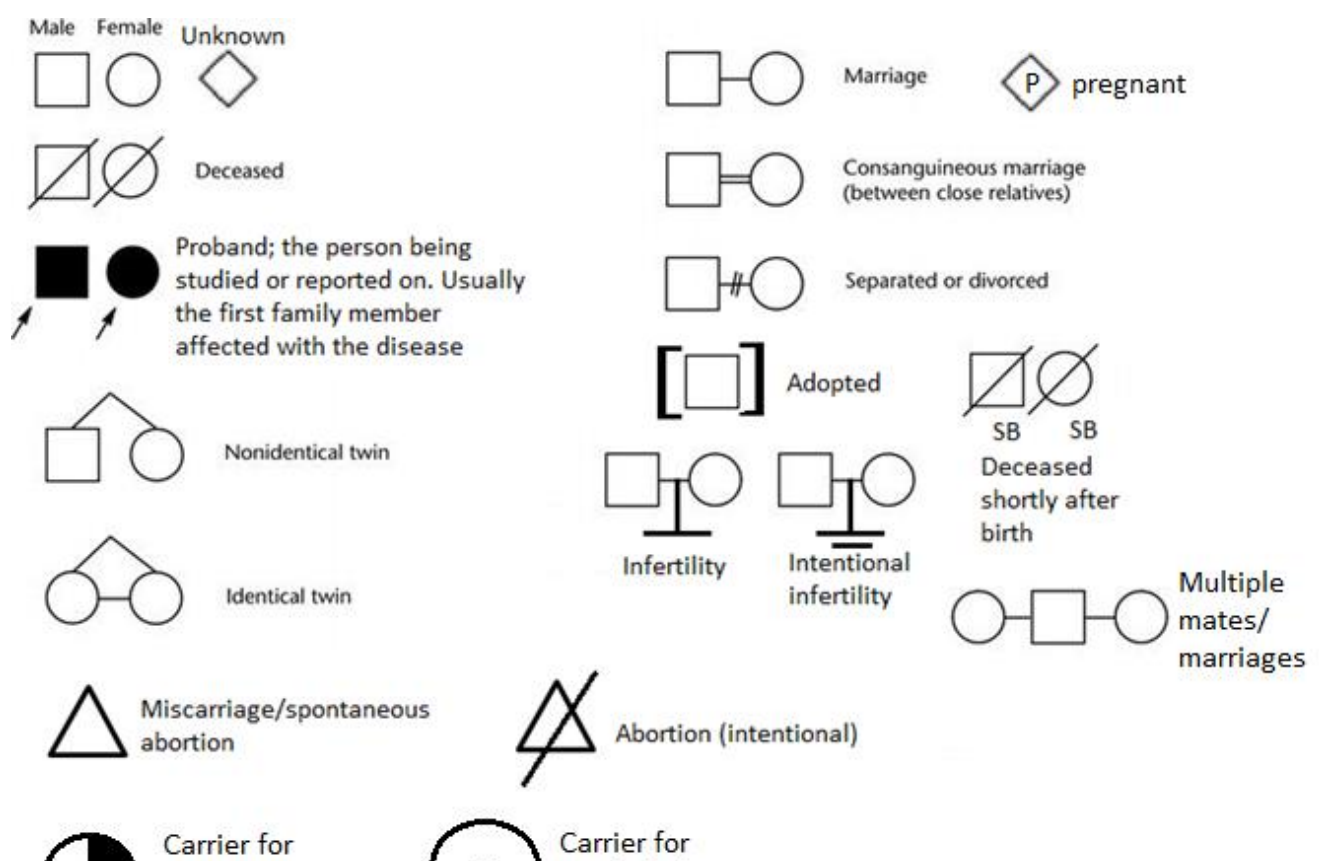
Polygenic inheritance and the environment: Some traits, such as size or height, weight, shape, color, metabolic rate and behavior are not the result of interaction between one, two, or even several genes; instead, they are the cumulative result of the combined effects of many genes. The phenomenon is known as *polygenic inheritance*. For example, males in the U.S.A are taller, on average, than they were 60 years ago, due to better nutrition and other environmental factors. Human skin color and intelligence are believed to be under the influence of a similar kind of genetic or environmental factors.

*in the polygenic inheritance of skin color, there are at least 4 genes affecting the production of melanin. For example, in the genotype aabbccdd, all the alleles are for low melanin production and the resulting

phenotype is an albino. On the other end of the spectrum, all the alleles in the genotype AABBCDD are for high-melanin production and so a person with this genotype has extremely dark skin. (Note: small letters → allele for low melanin production; Capital letters → alleles for high melanin production)

Patterns of inheritance: Many human traits are inherited. Hair and eye color, facial characteristics, and aspects of personality all are passed from parents to children. Some major clinical disorders are also inherited, and it is important for investigators to be able to study them, both to assess the likelihood that parents at risk will produce affected children and to seek future cures.

To study how a human trait is inherited, investigators look at the results of crosses that have already been made; they study family histories called *pedigrees*. By studying which relatives exhibit a trait, it is often possible to say if the gene producing the trait is sex-linked or autosomal and to determine if the trait's phenotype is dominant or recessive and which individuals are homozygous and which are heterozygous for the allele specifying the trait.



LG 3: Types of Genetic Diseases/Disorders

Each human is estimated to have approximately 20,600 different coding genes. Alterations in these genes, or recombinations of them, can produce genetic disorders. Genetic disorders are of the following types:

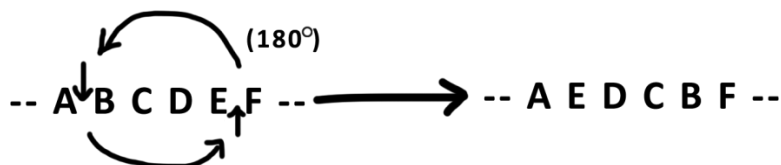
1. Chromosomal disorders.
2. Single gene disorders.
3. Multifactorial disorders.
4. Mitochondrial disorders.

1) Chromosome abnormalities (anomalies/aberrations): Certain genetic diseases are caused by alterations in the number or structure of chromosomes.

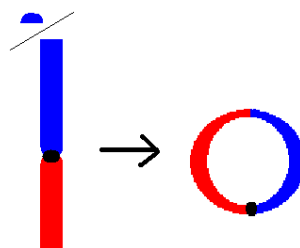
Usually because of mistakes during meiosis or mitosis, homologous chromosomes or their chromatids may not separate (non-disjunction) and the result is an abnormal number of chromosomes in the gamete produced and, later on, in the zygote. Numerical anomalies result in either **polyploidy**, the addition of complete haploid sets of chromosomes or **aneuploidy**, the addition or loss of one, two or, rarely, more chromosomes. Structural anomalies result from breakage and fusion of chromosome segments in a novel way (deletion, translocation, addition, inversion, duplication, ring chromosome, isochromosome, etc.). A deletion is simply the loss of a portion of a chromosome. A translocation occurs when a deleted portion of one chromosome is transferred to and becomes part of another, non-homologous chromosome.

*Different types of chromosomal mutations:

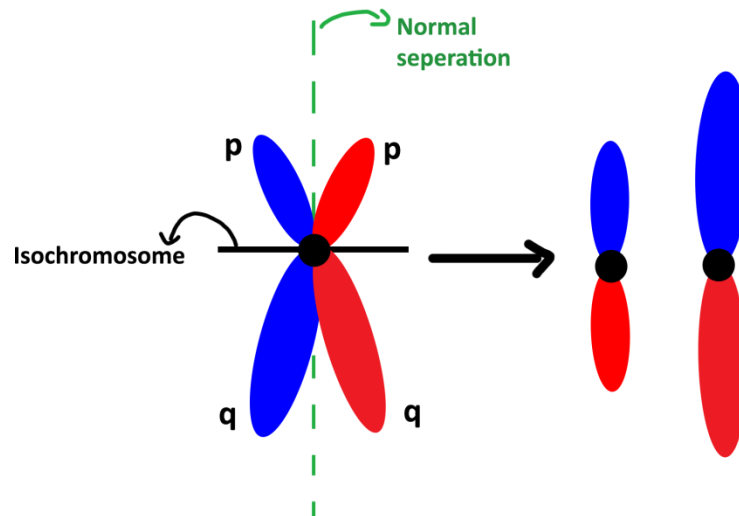
- Inversion mutation:



- Ring chromosome:



- Isochromosome:



The polyploidy conditions that have been observed in humans are triploidy (69; 3n or XXX) and tetraploidy (92; 4n or XXXX). The additional chromosomes encode a large amount of gene products, causing multiple anomalies such as defects of the heart and CNS. Triploidy and tetraploidy are both lethal. The most common cause of triploidy is the fertilization of an egg by two sperms (*dispermy*). Triploidy can also be caused by the fusion of an ovum and a polar body, followed by fertilization by a sperm. Tetraploidy can be caused by a mitotic failure in the early embryo.

*Polyploidy is lethal in humans but is useful in some plants and animals. In the liver, it occurs often due to high metabolic activity. Thus, polyploidy doesn't lead to the death of hepatocytes, as these cells can use the extra proteins and enzymes, due to said high metabolic activity.

Autosomal aneuploidy: Autosomal aneuploidy conditions consist primarily of monosomies and trisomies. They are usually caused by non-disjunction, the failure of chromosomes to disjoin normally during meiosis I or II. The resulting gamete either lacks a chromosome or has two copies of it, producing a monosomic or trisomic cell respectively. Autosomal monosomies are always lethal, while some autosomal trisomies are compatible with survival.

Trisomy 21 (Down syndrome)– 47, X X + 21 or 47, X Y +21:

Down syndrome (D.S), formerly called Mongolism (a term dropped by WHO in 1965), is seen in approximately 1/800 live births. Persons with D.S usually have three copies of chromosomes 21 in their karyotype. About 95% of cases are caused by non-disjunction during the formation of parental gametes (gametogenesis), the remainder being caused by chromosome translocation, most often resulting in chromosomes 14 and 21 being joined together.

D.S includes characteristics such as a particular facial appearance (round cheeks, folding of the eyelid (epicanthic fold), a low nasal root, and a flattened maxillary region), short stature, short fingers, a wide space between the 1st and 2nd toes, heart defects, abnormal teeth, a short neck, a protruding large tongue (a large tongue that makes the mouth remain open), and an increased susceptibility to infection and mental retardation (adults have the mental ability of a 9 year old, IQ is ~50-70). Another feature of D.S is hyper-mobile joint movement. The chance of developing leukemia is 20 times higher in persons with D.S.

The chances of a woman having a Down syndrome child increase rapidly with age, starting at about age forty. Although people with D.S, on average, have a lifespan much shorter than normal, some live to middle-age or beyond, even beyond 50 sometimes. In such cases, it's because 3% of DS individuals exhibit mosaicism, with some normal cells that have 46 chromosomes in addition to the ones with 47 chromosomes. The higher the percentage of cells with a normal set of chromosomes, the milder the DS phenotype. The mosaicism occurs because in some of the cells during embryogenesis, the extra 21st chromosome is degraded or lost.

Trisomy 18 (Edwards syndrome)– 47, X Y + 18:

This trisomy is the second most common autosomal trisomy, with a prevalence of about 1/6,000 live births. Trisomy 18 patients have prenatal growth deficiency, characteristic facial features (including small ears, a small mouth that is often hard to open, and a reduced chin), a short

sternum, hand abnormalities, congenital heart defects (like ventricular septal defects), a head that is elongated anteroposteriorly, congenital hip dislocation, and protrusion of the bowel into the umbilical cord (omphalocele). About 50% of children with trisomy 18 die in the 1st month, and only about 10% are still alive at 12 months of age.

Trisomy 13 (patau syndrome)– 47, X Y + 13:

This condition is seen in about 1/10,000 births. The malformation consists of an oral-facial cleft, microphthalmia (small eyes), polydactyly (extra fingers/toes) and an abnormal CNS. The survival rate is very similar to that of trisomy 18, with 90% of affected fetuses being spontaneously aborted. The risk of bearing a child with this condition increases with advanced maternal age.

Partial monosomy:

Cri du chat syndrome (46, 5p- ; a part of chromosome 5 is absent):

features include the glottis and larynx being abnormal (which results in the crying of the infants being high-pitched and sounding like the meowing of cat), gastrointestinal and heart complications, microcephalus and mental retardation.

Wolf-Hirschhorn syndrome (46, 4p-): Causes microcephalus, mental retardation, deafness and cardiovascular system abnormalities.

Sex chromosome aneuploidy: Non-disjunction may also produce individuals with an unusual number of sex chromosomes, but the consequences of this class of aneuploidy are less severe than those of autosomal aneuploidy.

Turner syndrome (45, XO):

Turner syndrome (T.S) is the only known viable human monosomy. About one girl in 3,000 has T.S. These individuals are phenotypically female, their sex organs do not mature at adolescence, and secondary sex characteristics fail to develop. Such individuals are sterile (they do not

undergo puberty or menstruate), though a mosaic individual may be fertile and be able to give birth. Girls with T.S exhibit short stature, have a broad chest and small breasts, a low hairline, a triangular face and a webbed neck. Many patients with T.S have congenital heart defects. There is typically some diminution in spatial perceptual ability, but females with T.S are not mentally retarded.

*Mosaic/mosaicism is the presence of two or more populations of cells with different genotypes in one individual who was developed from a single fertilized egg.

Klinefelter syndrome (47, XXY):

About one young man in 1,000 male births is found to have K.S. These men tend to be taller than average, with long arms, legs and fingers and they are usually not mentally retarded. Affected individuals are sterile males, their testes are under-developed (and small), have reduced testosterone amounts, reduced masculinization and there may be some breast development as well as more rounded hips. In addition, they have reduced facial and pubic hair and reduced hair in other locations. These phenotypic abnormalities are not apparent until puberty. It is estimated that at least half of "47, XXY" conceptions are spontaneously aborted. Individuals with the 48, XXXY and 49, XXXXY karyotypes have also been reported. Because they have a Y chromosome, they still have a male phenotype, but the degree of mental deficiency and physical abnormalities increase with each additional X chromosome.

47, XYY syndrome (Jacob's syndrome):

Males with this karyotype tend to be taller than average, suffer from persistent acne, and tend to have barely normal intelligence. In some cases, the person may also have problems with language and speech, as well as learning difficulties. At one time, it was suggested that these men were likely to be criminally aggressive, but it has since been shown that the incidence of such violent behavior among them is no greater than among XY males.

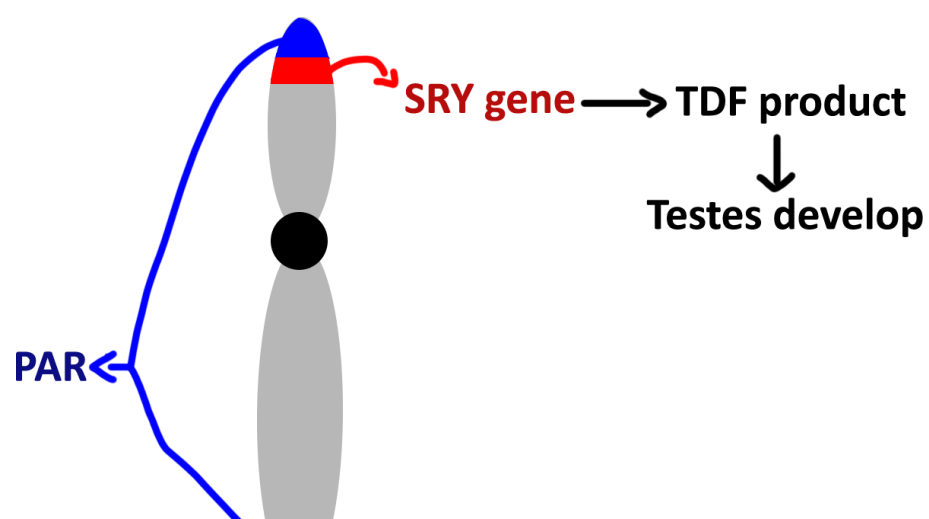
Trisomy (Meta/Super female)– 47, XXX:

This karyotype occurs in 1/1,000 females and usually has benign consequences. In some cases, there is a tendency toward physical abnormalities; sometimes the person suffers from sterility, menstrual irregularity, or mild mental retardation. They are also taller than average and tend to have cardiovascular issues. Females have also been seen with 4, 5 or even more X chromosomes. Each additional X chromosome is accompanied by increased mental retardation and physical abnormalities.

Note: Barr bodies = No. X chromosomes - 1

XX males and XY females:

During normal meiosis in the male, crossover occurs between the tip of the short (p) arm of the Y chromosome and the tip of the short arm of the X chromosome. These regions of X and Y chromosomes contain highly similar DNA sequences. The distal portion of the Y chromosome is known as the *pseudo-autosomal region (PAR)*. Occasionally, the crossover occurs just outside the pseudo-autosomal region, where lies a gene known as SRY (sex-determining region on the Y). This gene, which is expressed in embryonic development, encodes a product that interacts with other genes to initiate the development of the undifferentiated embryo into a male. As a result of a faulty crossover between the X and Y chromosomes, such that the SRY gene, instead of remaining on the Y chromosome, is transferred to the X chromosome, an offspring receiving this X chromosome will be an XX male, and an offspring receiving the Y chromosome that lacks the SRY gene would be an XY female.



LG 4:

Autosomal partly dominant inheritance: Sickle cell anemia is partly dominant. Sickle cell anemia (SCA) is an example of a human disorder that is controlled by an incompletely dominant allele. SCA occurs at a high rate among blacks. In persons with sickle cell trait (heterozygous; $Hb^A Hb^S$), the hemoglobin contains both normal and sickle cell β -polypeptide chains. They do not usually have any sickle-shaped cells unless they experience dehydration or mild oxygen deprivation (when subjected to extreme exercise, such persons are more likely to die). Persons with sickle cell disease (homozygous; $Hb^S Hb^S$) have irregular or sickle shaped RBCs. The defect is caused by an abnormal hemoglobin that accumulates inside the cells.

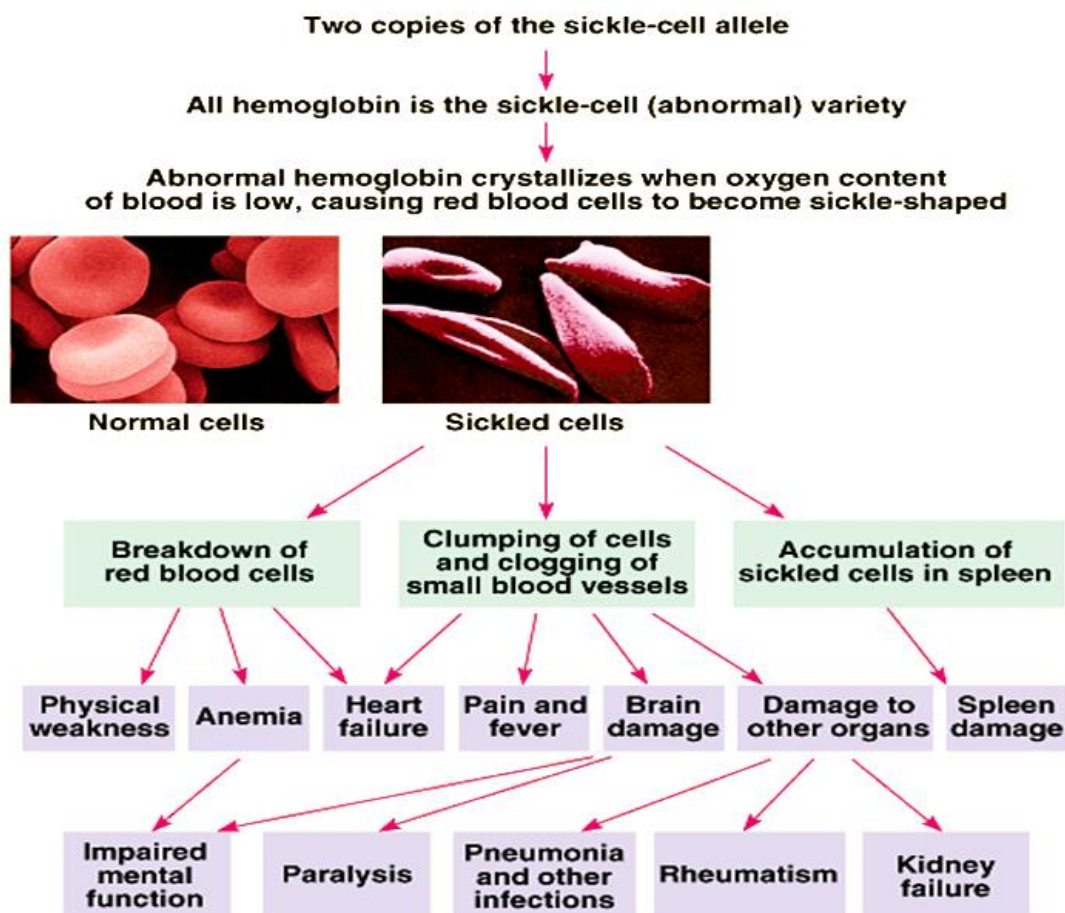
*Adult hemoglobin contains α - and β -globin. In people with SCA, the β -globin is malformed; this is caused by a single nucleotide substitution (in which GAG is converted into GUG), resulting in a glutamate residue being replaced by valine. This is normally a benign mutation, causing no apparent effects on the secondary, tertiary, or quaternary structures of hemoglobin in conditions of normal oxygen concentration. However, under low oxygen concentration, Hb S polymerizes and forms fibrous precipitates. Glutamate and valine are two different types of amino acids (the former is acidic and polar and the latter is nonpolar), and this is why this single amino acid substitution has such a drastic effect.

Note: Normal RBCs are flexible (allowing them to pass through capillaries), have a bi-concave shape and have life spans of 90-120 days. Sickling RBCs, on the other hand, are more rigid and less flexible, have the shape of a sickle and have life spans of 10-20 days (as they are more fragile).

Because sickle shaped cells are less flexible and are unable to pass along narrow capillaries like disk-shaped cells (normal RBCs) can, they clog the vessels and break down. This results in poor circulation due to vascular obstruction that produces localized hypoxemia and infarctions of various

internal organs, including bones, kidneys and lungs. Premature destruction of the sickle erythrocytes produces anemia and jaundice.

*As mentioned, sickle cell anemia is common among black individuals. In people with sickle cell trait, a benefit of sickle-shaped RBCs is that they provide some resistance against malaria, as the parasite can't infect the abnormally-shaped RBCs. This gives those with sickle cell trait a selective advantage compared to those who don't in environments where malaria is prominent (e.g., Africa), hence why this trait has spread among black people.



Signs and symptoms of SCA include yellow skin, pain (especially in muscles and bones, due to hypoxia), fatigue, bone deformities, feeling cold, brittle and thin bones (more prone to fractures), splenomegaly and stroke, which is the main cause of death. Extramedullary hematopoiesis- hematopoiesis (blood cell production) occurring in organs outside of the bone marrow- can also result.

*anemia happens as result of hemolysis

*One of the functions of the spleen is filtering pathogens out of blood. So, the spleen damage that results from SCA causes a decrease in immune ability and leaves those with the condition more susceptible to infections

Medical management of SCA:

- Vaccination against *Streptococcus pneumoniae* (these vaccines are called pneumococcal vaccines)
- Antibiotics, as SCA individuals are more susceptible to infection due to spleen dysfunction
- Bone marrow transplantation
- Drugs, e.g., hydroxyurea, which up-regulates γ -globin synthesis
- Rehydration (giving water intravenously, prevents acidosis)
- Dietary folic acid supplements (folic acid helps make new RBCs; it's a precursor for DNA synthesis)
- Gene therapy: stem cell transformation

Extra-nuclear inheritance (mitochondrial inheritance, also called maternal or Eve inheritance): Each human cell contains a several hundred or more mitochondria dispersed throughout the cytoplasm and they have their own DNA molecules, occurring in several copies per organelle. The mitochondrial DNA (mtDNA) is circular and encodes 37 genes; two ribosomal RNAs, 22 tRNAs and 13 polypeptides involved in oxidative phosphorylation to produce ATP. Mitochondria are still considered only semi-independent, as ~90 of their proteins are coded for by nuclear DNA.

As mentioned, mitochondrial enzymes are usually encoded by nuclear genes, but some of the cytochrome oxidase enzymes are derived solely from mitochondria. Because it is located in the cytoplasm, mtDNA is inherited exclusively through the maternal line. Males cannot pass it to their offspring because sperm cells contain only a few mtDNA molecules that do not enter the egg.

Mutations involving the mtDNA are known to account for a small number of genetic diseases. One of the best known mitochondrial diseases is Leber Hereditary Optic Neuropathy (LHON; a late-onset disease). This disease is characterized by rapid loss of vision (bilateral) as a result of optic nerve death. Vision loss typically begins in the 3rd decade of life and is irreversible.

MERRF syndrome (Myoclonic Epilepsy with Ragged Red Fibers) is a mitochondrial disease. In muscle biopsy, ragged red fibers can be seen, hence the disease's name. Symptoms of this condition include ataxia (loss of muscular coordination), deafness, blindness, hair loss, dementia, myopathies, and epileptic seizures. In MERRF syndrome, the more energy-consuming organs, such as the brain, are most affected. Yet another example of a mitochondrial disease is Kearns-Sayre syndrome (KSS), which is characterized by blindness, deafness, dementia, droopy eyes, heart conditions, myopathies and paralysis of eye muscles.

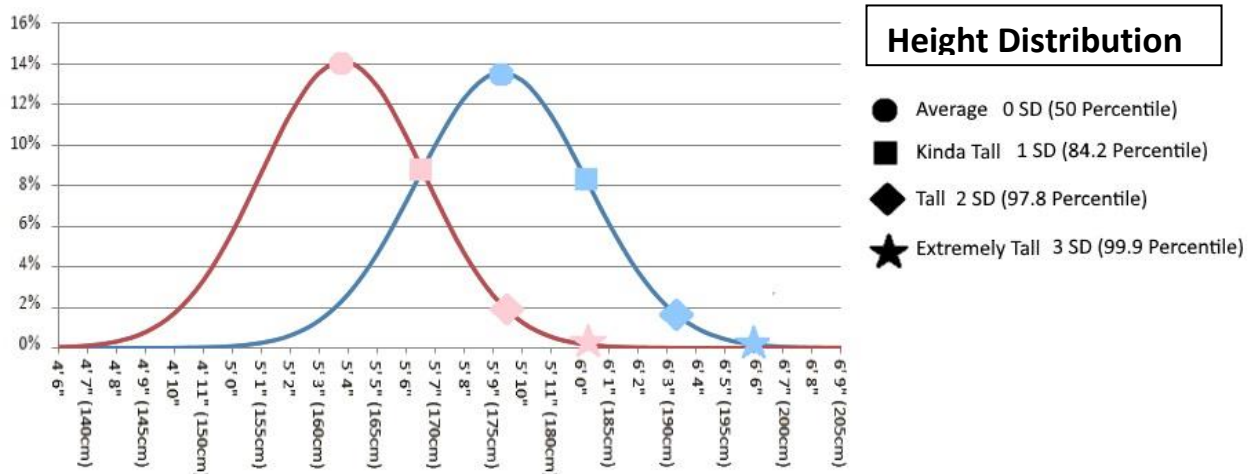
*It has been suggested that mtDNA mutations, which accumulate throughout the life of the individual as a result of free radical formation, could contribute to the aging process. Free radicals have the ability to cause mutations. High caloric intake and heavy exercise induce mutations as they increase the production of these free radicals. mtDNA has no repair mechanisms and is more susceptible to mutations due to the high concentration of free radicals.

Note: 2 factors that contribute to aging are mtDNA oxidative damage accumulation and telomere shortening

Multifactorial inheritance and common diseases (disorders):

A trait being “multifactorial” means that both genetic and environmental factors collectively influence phenotype. For example, a child of tall parents is more likely to be taller than is a child of short parents, but poor nutrition (an environmental factor) may compromise the growth of either child. Most quantitative phenotype variations among normal individuals are due to multifactorial traits. Examples include the range of skin

pigmentation, weight, height and intelligence. These traits are said to have a continuous variation and their quantities and distribution produce a normal distribution curve.



Many human diseases are also inherited as multifactorial traits. The long list of Multifactorial diseases includes cleft lip and/or palate, clubfoot, congenital dislocation of the hips, hypertension, heart disease, atherosclerosis, diabetes mellitus, allergies, cancers, obesity, some neurodegenerative diseases (such as Parkinson's and Alzheimer's disease) and some forms of mental illness, such as schizophrenia and manic-depressive psychosis. Many genes affect our cardiovascular health, making some of us more prone than others to heart attacks and strokes. However, a healthy diet, exercise, a lack of smoking and an ability to put stressful situations in perspective all reduce our risk of heart disease.

LG 5: Prenatal Genetic Testing (screening and diagnosis of genetic diseases)

Genetic screening: The process of identifying persons at risk of producing children with genetic defects and assessing the genetic state of early embryos and neonates.

Types of prenatal test: Today, many prenatal tests are available to detect chromosomal and gene mutation. Karyotyping* of fetal or embryonic cells can indicate whether a developing child has Down syndrome or one of the other chromosomal abnormalities. Biochemical tests are available for dozens of enzymes such as hexosaminidase A, the missing enzyme in persons with Tay-sachs disease. (*A karyotype is a “portrait” of the chromosomes during mitosis metaphase)

Some prenatal tests are screening tests, while others are diagnostic. A screening test gives information about the possibility that a problem exists and is presumptive. For example, an α -fetoprotein (AFP) screening test can show that there is a 1% risk of Neural tube defects. A diagnostic test can tell that a specific problem is present with certainty. An example of such a test is amniocentesis, which is offered to confirm or exclude the presence of trisomy 21 and other chromosome abnormalities.

Diagnostic prenatal testing can be done by **invasive** or **non-invasive** methods (and there are some tests that are **less invasive** than tests considered to be invasive, and these involve taking a drop of blood from the mother).

An invasive method involves probes, needles or catheters being inserted into the amniotic sac, placenta or umbilical cord. Examples include amniocentesis, chorionic villus sampling and fetal blood sampling. Non-invasive techniques include examinations of the woman's womb through ultrasonography and maternal serum screening (i.e., screening alpha-fetoprotein, fetal cells and cell-free fetal DNA in maternal blood) (maternal serum screening is an example of a less invasive test).

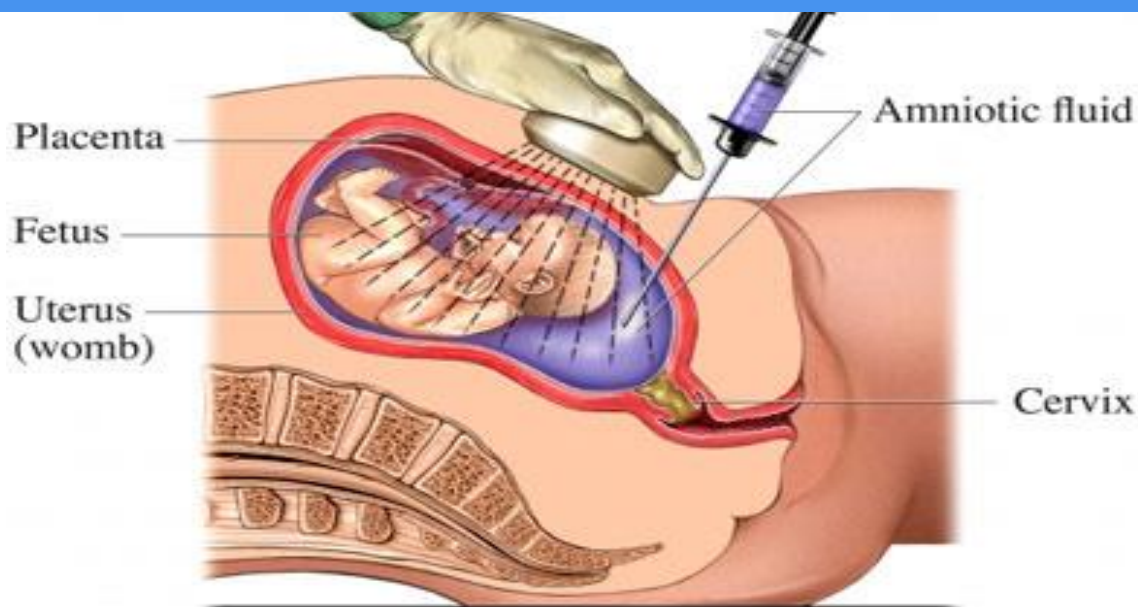
Reasons for prenatal diagnosis:

- 1- To enable timely medical or surgical treatment of a condition before or after birth.
- 2- To give the parents the chance to abort a fetus with the diagnosed diseased condition.
- 3- To give parents the chance to prepare psychologically, socially, financially and medically for a baby with a health problem or disability or for the likelihood of a stillbirth (death or loss of a baby before or during delivery).

Who should consider having a prenatal gen. diagnostic test?

- 1- Mothers older than 34 years old.
- 2- Mothers whose ethnic background has been associated with certain types of genetic disorders such as Tay-Sachs disease (for Ashkenazi Jewish), sickle cell anemia (for those with an African American background) or cystic fibrosis (for those with a Caucasian background).
- 3- Mothers who have a family history or who have had a previous child with a genetic condition.
- 4- Mothers who have experienced multiple miscarriages.
- 5- Mothers who have had a prenatal screening test that showed they were at increased risk of having a baby with a chromosomal abnormality such as DS, trisomy 18 or spina bifida.
- 6- Couples who are also close relatives (e.g., first cousins), since there is a higher chance of both being carriers for the same recessive condition and hence a higher chance of the child having said recessive condition.
- 7- Pregnant women who have been exposed to environmental factors that could cause diseases in the child (mutagen and teratogens), e.g., radiation, drugs and viral infections.

Amniocentesis: In the 4th month of pregnancy (i.e., 2nd trimester), a small sample of amniotic fluid is removed by means of a sterile needle. A needle is inserted through the abdominal wall in to the amniotic sac after ultrasound localizes the placenta and determines the position of the fetus.

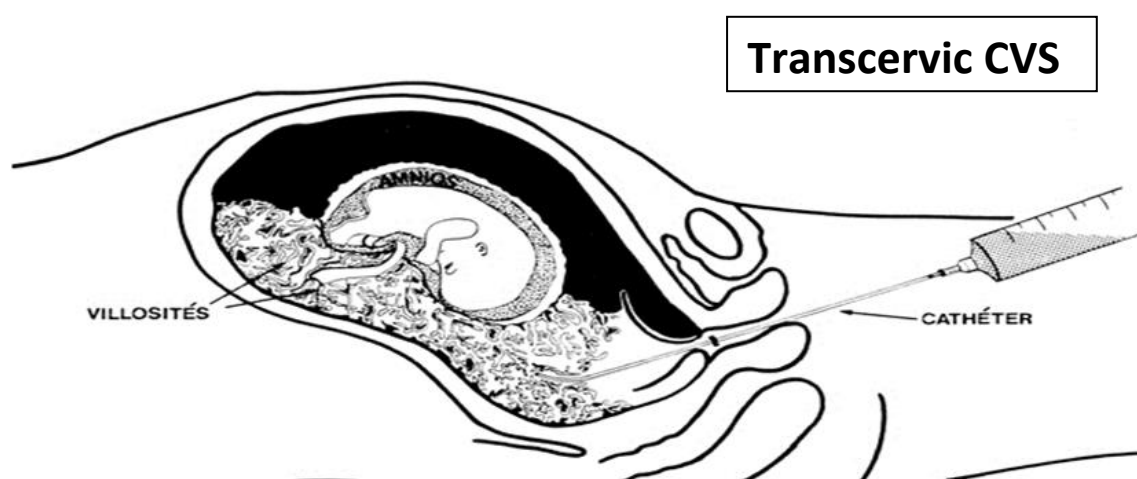


About 20-30 ml of amniotic fluid is withdrawn; this fluid contains living cells (called *amniocytes*) shed by the fetus (and hence containing the genetic and chromosomal makeup of the fetus). These cells can be grown in tissue cultures and multiply into adequate numbers for testing purpose (by Karyotyping, biochemical tests and DNA analyses).

A disadvantage of amniocentesis is that it requires time so that the amniocytes are cultured and there is also a risk of infection developing as the puncture may cause leakage. Its advantage, on the other hand, is that it's safer than other methods (i.e., has a lower chance of causing abortion).

Chorionic villus sampling (CVS, also called placenta biopsy):

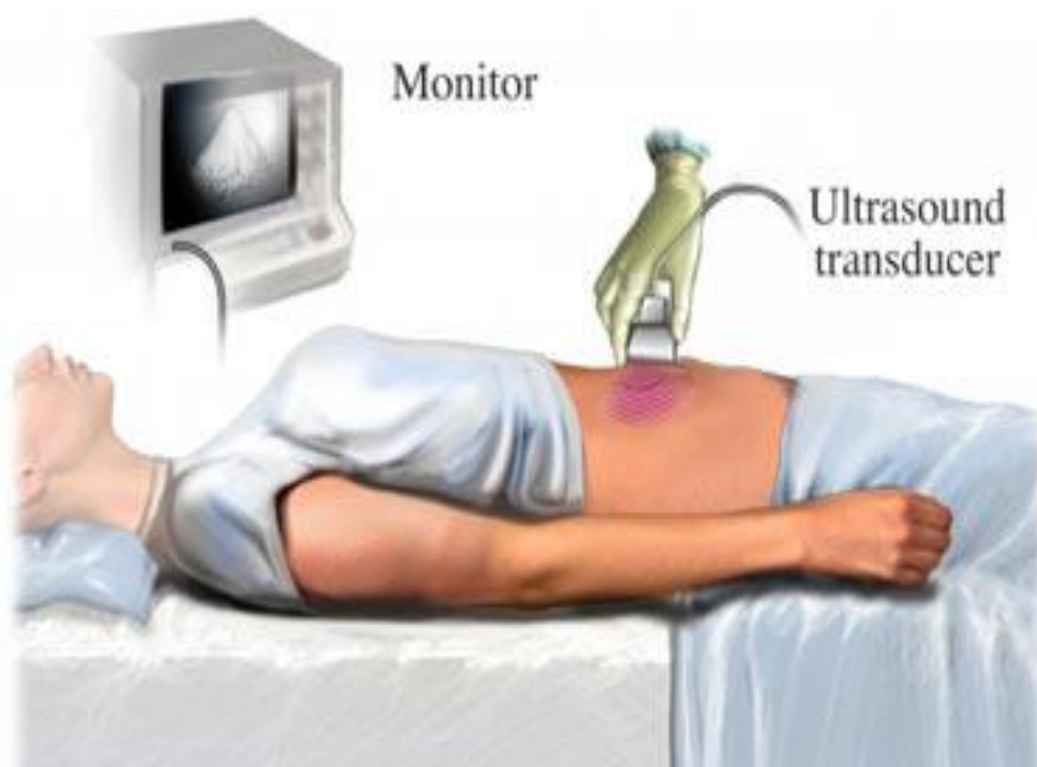
CVS is performed by aspirating fetal trophoblastic tissue (chorionic villi) by either a trans-cervical or trans-abdominal approach. It is usually performed at 10 weeks of pregnancy (in the 1st trimester). The chorionic villi contain cells with the same chromosomal and genetic makeup as the fetus and, therefore, these cells can be tested for possible fetal abnormalities.



The advantage of CVS is that testing can be done much earlier (compared to amniocentesis, for example) and hence an abortion, if desired, can be done during the 1st trimester when it is considered medically safer for the mother. CVS also takes less time, since there is no need for culturing. Disadvantages of this technique include the possibility of CVS causing reduced limbs and the risk of abortion. In addition, samples taken by CVS can be contaminated by maternal cells, and hence seem to exhibit mosaicism and give false results.

*Amniocentesis samples can also show mosaicism due to mutations during culturing.

Ultrasonography: is the most common technique now used for fetal visualization. It is used to confirm a pregnancy and to test for specific congenital malformations, e.g., a short limb skeletal dysplasia (dwarfism), anencephaly, macrocephaly, spina bifida and poor fetal growth. Sometimes it can be used to evaluate the uncertain gestational age, amniotic fluid volume and determine the baby's sex. Down syndrome can be diagnosed with ultrasonography by checking for an underdeveloped or absent nose bone, or by nuchal translucency scan (this test assesses the quantity of fluid collecting within the nape of the fetal neck; fetuses with Down syndrome have increased nuchal translucency).



Maternal blood sampling: Some genetic conditions can be detected by mother serum screening. For instance, we can check levels of important markers produced by the fetus and placenta (e.g., alpha-fetoprotein (AFP), estriol, β -hCG and inhibin-A) in the woman's serum. The level of alpha-fetoprotein is significantly higher than normal when the fetus has neural-tube defects or may be decreased in the case of Down syndrome.

*AFP levels are increased in women pregnant with twins or triplets, so in the case of high AFP levels, a differential diagnosis is needed to avoid misdiagnosis. AFP level testing can't really be considered diagnostic.

* AFP is a major plasma protein produced by the yolk sac and the fetal liver during fetal development. It is thought to be the fetal analog of serum albumin. The function of AFP in adult humans is unknown; however, in some animals, it binds estradiol to prevent the transport of this hormone across the placenta to the fetus. The main function of this is to prevent the masculinization of female fetuses.

Fetal cell sorting: During pregnancy, a few fetal cells are released into the mother's circulatory system. Recent advances have made it possible to separate fetal cells from a maternal blood sample (automated cell-sorting machines; fluorescence-activated cell sorting by laser guidance*). Fetal cells can be detected and separated from maternal blood. The fetal cells obtained can be cultured for chromosome analysis or DNA molecular testing.

*Every individual has unique antigens. Fluorescent antibodies attach to these antigens and the cells are then separated by laser guidance.

LG 6: Treatment of Genetic Diseases

Gene therapy: Gene therapy techniques involve the replacement of a non-functional or missing gene product by inserting a normal gene into a somatic cell (Hepatocyte, Lymphocyte, Skin fibroblast, brain cells, skeletal muscle cells, stem cells, etc). There are many physical, chemical, and biological methods being used for introducing genes into cells; physical methods include microinjection, electroporation, particle bombardment (also called biolistics, uses a gene gun) and chemical methods include liposomes that carry DNA across the plasma membrane and the use of calcium phosphate or calcium chloride, which disturbs the cell membrane and allows negative DNA to enter the cell (it allows it to overcome the repulsion it has with the negatively charged phosphate groups of the cell membrane's phospholipids). Biological approaches to gene transfer use a vector, such as a viral genome (genetically modified viruses). Some viruses, such as Retroviruses and Adenoviruses, are able to insert copies of normal human gene in to the affected cells. Examples of viral vectors include Adeno associated virus and lentivirus (complex retroviruses, used for the treatment of HIV infections).

Note: Retroviruses target dividing cells (e.g., stem cells). Adenoviruses target non-dividing cells, and are used for the treatment of hemophilia B (factor IX deficiency), also called Christmas hemophilia, as well as cystic fibrosis. Adeno associated viruses target non-dividing cells, and carry a small amount of DNA (~5kbp). Adeno associated viral vectors can be used for hemophilia B treatment.

The transplantation of cells, tissue or organs, e.g., liver transplantation and bone marrow transplantation can be considered as a form gene therapy because the cells transferred to the recipient continue to function like the donor genome. An example is cystic fibrosis, for which lung transplantation is a possible treatment.

The replacement of a deficient substance may be a possible mode of treatment, e.g., in classical Hemophilia (Hemophilia A; factor VIII deficiency), the replacement of the deficient clotting factors VIII correct the bleeding disorder. Another example is type I insulin-dependent diabetes mellitus, which is treated by the administration of insulin.

*another form of gene therapy is anti-sense gene therapy (used to treat Huntington disease)

*gene therapy can be used for the treatment of both genetic and non-genetic disorders

There are two main types of gene therapy:

- 1- somatic cell gene therapy
- 2- reproductive or germ-line gene therapy

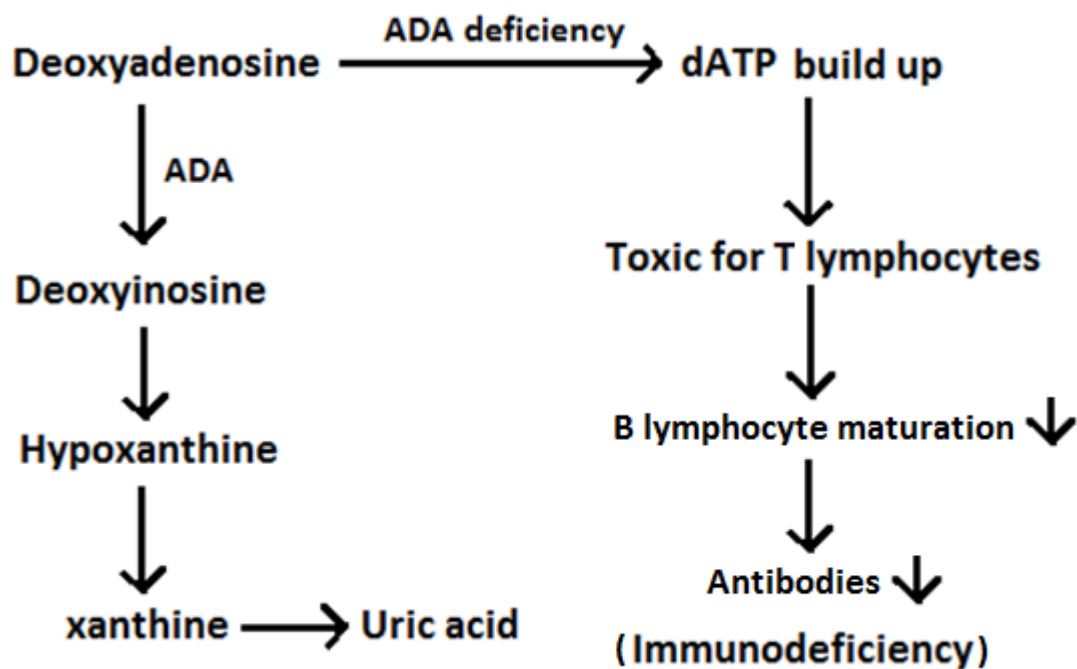
Germ-line cell therapy involves the introduction of corrective genes into reproductive cells (sperm and eggs) or zygotes, with the objective of creating a beneficial genetic change that is transmitted to the offspring. Disadvantages of this method include the fact that it's prohibited due to concerns over ethics and safety, as it carries a high chance of malignancy, causes sperms and eggs to die, and the gene is randomly inserted and may cause malformations to significant genes (e.g., tumor suppressor genes and proto-oncogenes). Its advantage is that the mutation it corrects is corrected for the offspring's whole life (i.e., its effect is permanent).

*A proto-oncogene is a normal gene which, when altered by mutation, becomes an oncogene, a gene that has the potential to cause cancer.

Gene therapy of somatic cells, those not directly related to reproduction (ova and sperms), results in changes that are not transmitted to the offspring. An example of gene therapy in somatic cells is the introduction of genes in an organ or tissue to induce the production of an enzyme, e.g., ADA (Adenosine Deaminase). A disadvantage of somatic cell gene therapy is that its effects are temporary.

The first successful gene therapy was done in 1990 on Ashanti DeSilva, who had a form of severe combined immunodeficiency (SCID) caused by ADA deficiency. The gene therapy involved extraction T-cells and transforming them with viral vectors, and then injecting the treated cells back into the patient.

Note: ADA enzyme function



*ADA deficiency causes susceptibility to viral, bacterial and fungal infections.